Assignment 3: Finoptix

Report

-By Vandit Gupta (241132)

In this assignment, we used the 3-Factor Fama French model to estimate market returns and apply mean-variance optimization on a portfolio to obtain a portfolio with the best shape ratio.

Interpretation of Betas:

HML: A Positive value indicates that the stock performs similar to a growth stock, while a negative value suggests that the stock performs similar to a value stock.

SMB: Higher beta – Behavior like small cap stock

Lower- Beta – Behavior like High Cap Stock

Market-RF: A value of more than 1 indicated that the stock is more volatile than the market, whereas a value less than 1 indicated less sensitivity to market moves.

Optimization Results. (For Tickers of the top 15 Stocks)

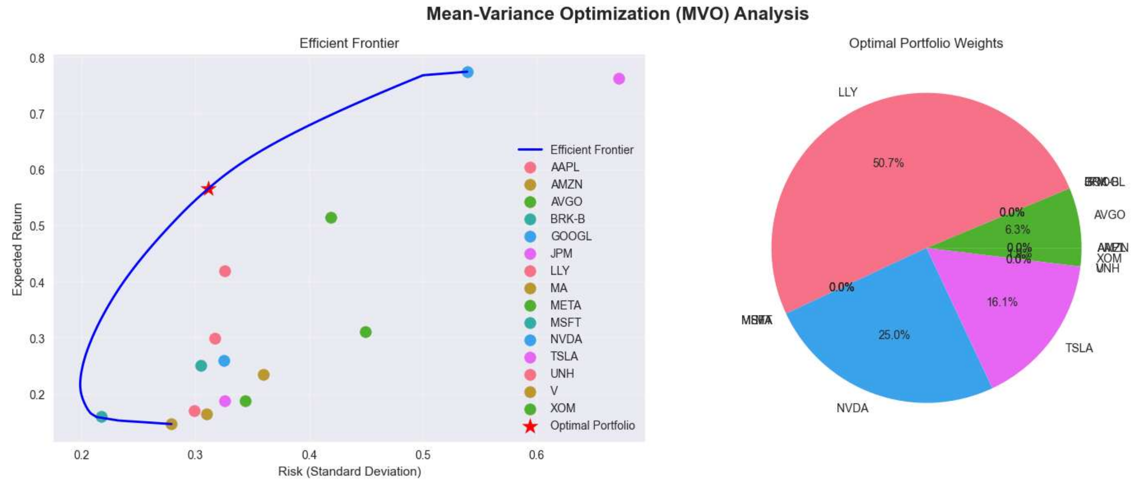
Tickers -- "AAPL", "MSFT", "GOOGL", "AMZN", "NVDA", "META", "BRK-B", "TSLA", "LLY", "JPM", "V", "UNH", "XOM", "MA", "AVGO"

Historical Average-Based Portfolio

Expected Return: 0.5651 (56.51%)

Risk (Std Dev): 0.3110 (31.10%)

Sharpe Ratio: 1.7526

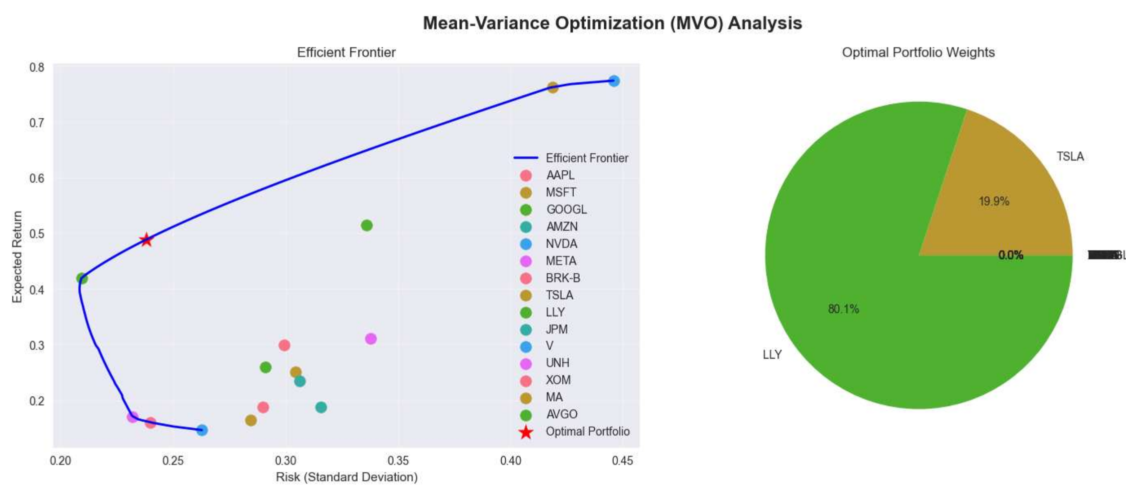


Famma French Based Porfolio

Expected Return: 0.4881 (48.81%)

Risk (Std Dev): 0.2379 (23.79%)

Sharpe Ratio: 2.0128



Results:

I feel the algorithm provided a safer model in the Fama-French Model, giving away some profit. Also, by looking at the line graph in the notebooks, we can see the irregularities in the actual returns and the returns calculated using the betas, which shows that a poor regression model was used.